

LESSON 16

MODULAR ARITHMETIC

Overview:

This lesson will introduce students to modular arithmetic and its applications. The lesson provides opportunities to apply and extend ideas to unfamiliar contexts. Students will begin with the familiar problems involving an analog clock and will soon realize that they already have some experience with modular arithmetic. Using this knowledge, mathematical ideas and notations are developed to help students solve problem involving basic number theory. Applications to serial number coding are also discussed.

Learning Outcomes:

At the end of this lesson, the students can:

1. perform basic operations in modular (clock) arithmetic;
2. encode and decode messages using simple shift ciphers; and
3. apply modular arithmetic to real-life contexts.

Materials Needed:

- Module (Printed/Softcopy)
- Internet Connection
- Smartphone/Laptop, Google Classroom, FB Group Chat and YouTube videos.

Duration: 5 hours

Learning Contents:

16.1 MODULAR ARITHMETIC

Modular arithmetic is a method for finding remainders where all the possible numbers (the numbers less than the divisor) are put in a circle and then counting around the circle the number of times of the number being divided, the remainder will be the final number landed on.

Clock arithmetic can be used to introduce modular arithmetic. Consider arithmetic that only uses the numbers on a clock except that 0 replaces 12. The set of numbers for the clock arithmetic is: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$.



Clock arithmetic refers to the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$ and the rules for adding, subtracting, multiplying, and dividing these numbers. The symbols for clock arithmetic are $+$, $-$, $\times$, and $\div$.

What is 18 in clock arithmetic? It can not be 18 since there is no number 18 . From now on, write 18 in mod 12 as 6 to distinguish it from 18 in base 10.

EXAMPLE 16.1:

Solution:

If I start 9 on the clock and move 9 hours clockwise, then I will end up at 6, therefore,

EXAMPLE 16.2: Suppose it is now Friday (5th day), what day is 8 days earlier?

Solution:

Modular subtraction can be defined in terms of addition. If Friday is the 5th day and we want to know what day is 8 days earlier. That is, which means we start at 5 and count back 8. We would end up 4 and 4th day is Thursday. Thus, 8 days earlier than Friday is Thursday.

EXAMPLE 16.3: Compute: (a) $40 \pmod{12}$ in clock arithmetic.

Solution:

a) or using clock, Go 40 hours clockwise from 12 and you will end up at 4.

b) However, the multiples of 3 in clock arithmetic are 3, 6, 9, and 12 and each of these multiple fails to give a remainder of 2 when divided by 3.

16.2 CONGRUENCE MOD n

For integers a , b is congruent to c modulo n , written $a \equiv b \pmod{n}$ where n is a natural number, if and only if $a - b$ is an integer. The number n is called **modulus** and the statement $a \equiv b \pmod{n}$ is called **congruence**.

EXAMPLE 16.4: Determine whether the congruence $12 \equiv 0 \pmod{12}$ is true.

Solution:

and 12 is an integer. Hence, $12 \equiv 0 \pmod{12}$ is true.

EXAMPLE 16.5: April 25, 2019 was Thursday. What day of the week is April 25, 2025?

Solution:

There are 365 days in a year and 366 days in a leap year. From the problem, there are six years between the two dates, where two of these are leap years. Thus, the total number of days is given by $6 \times 365 + 2 \times 366 = 2192$. Because remainder of 1 or by definition, So the day of the week 2192 days after April 25, 2019 will be the same as the day 1 day after April 25, 2019. Thus, April 25, 2025 will be Friday.

EXAMPLE 16.6: Disregarding A.M. or P.M., if it is 5 o'clock now, what time was 57 hours ago?

Solution:

The time can be determined by calculating $-57 \pmod{12}$. Since -57 is a negative number, we have to find a whole number less than the modulus 12, such that $-57 + 12k$ is an integer. However, only 8 and is less than 12 can satisfy the definition, that is when $k=5$ is an integer. Thus $-57 + 12 \times 5 = 3$. Therefore, if it is 5 o'clock now, 57 hours ago was 8 o'clock.

16.3 APPLICATION OF MODULAR ARITHMETIC

I. ISBN (International Standard Book Number)

Every book that is cataloged in the Library of Congress of must have an ISBN. This 13-digit number was created to help ensures that the order of books is filed accurately and that books are cataloged correctly.

The first three digits of an ISBN are 978 (or 979), followed by digits that are divided into three groups of various length. These indicate the country or region, the publisher, and the title of the book. The last digit (the 13th one) is called the **checked digit**.

Formula for the ISBN Checked Digit

$$d_{13} = 10 - (d_1 + 3d_2 + d_3 + 3d_4 + d_5 + 3d_6 + d_7 + 3d_8 + d_9 + 3d_{10} + d_{11} + 3d_{12}) \pmod{10},$$

if $d_{13} = 10$, then the checked digit is 0.

EXAMPLE 16.7: Determine whether the check digit of the ISBN for the Philippine edition of the Mathematics in the modern world is correct or not.
ISBN 978 - 971 - 23 - 9357 - 0

Solution:

$$d_{13} = 10 - (9 + 3(7) + 8 + 3(9) + 7 + 3(3) + 9 + 3(3) + 5 + 3(7)) \pmod{10}$$

$$d_{13} = 10 - 130 \bmod 10$$

$$d_{13} = 10$$

Since $d_{13} = 13$, hence the check digit is 0. Therefore, the ISBN has a correct digit.

II. UPC (Universal Product Code)

The universal product code is a number place on many items and is particularly useful in grocery stores (see figure at the right). The UPC is a12-digit number that satisfies a modular equation that is similar to the one for ISBNs. The last digit is the check digit.



Formula for the UPC Check Digit

$$d_{13} = 10 - (3d_1 + d_2 + 3d_3 + d_4 + 3d_5 + d_6 + 3d_7 + d_8 + 3d_9 + d_{10} + 3d_{11}) \bmod 10,$$

If $d_{13} = 10$, then the checked digit is 0.

EXAMPLE 16.8: Find the check digit of the Blu-ray Disc release of the film *Jurassic World*. The first eleven digits are 0-25192-21221- ____ .

Solution:

$$d_{13} = 10 - [3(0) + 2 + 3(5) + 1 + 3(9) + 2 + 3(2) + 1 + 3(2) + 2 + 3(1)] \bmod 10$$

$$d_{13} = 10 - (0 + 2 + 15 + 1 + 27 + 2 + 6 + 1 + 6 + 2 + 3) \bmod 10$$

$$d_{13} = 10 - 65 \bmod 10$$

$$d_{13} = 5$$

Hence, the check digit is 5.

III. Credit Card Numbers

Credit card numbers are normally 13 to 16 digits long. The first one to six digits are used to identify the card issuer. The table below shows some of the identification prefixes used by four popular card issuers.

Card Issuer	Prefix	Number of Digits
Master Card	51 to 55	16
Visa	4	13 or 16
American Express	34 to 37	15
Discover	6011	16



The primary coding method is based on the *Luhn algorithm*, which uses mode 10 arithmetic.

EXAMPLE 16.9: Determine whether 5234 8213 3410 1298 is a valid credit card number.

Solution:

Highlight every other digit, beginning with the next-to-last digit and reading from right to left.

5 2 3 4 8 2 1 3 3 4 1 0 1 2 9 8

Next, double each of the highlighted digit.

10 2 6 4 16 2 2 3 6 4 2 0 2 2 18 8

Finally, add all digits, treating two-digit numbers as two single digits.

Since , this is a valid credit card number.

IV. Cryptology

Cryptology is a study of making and breaking of secret codes. **Plaintext** is the actual undisguised message (usually an English message) that we want to send. On the other hand, the secret disguised message that is transmitted is called **ciphertext**. **Encryption** (encipherment) is the process of converting plain text to ciphertext and the process of converting ciphertext back to plain text is called **Decryption** (decipherment).

Notation: Z_n represents all possible remainders in a system, that is,

$$Z_n = \{0, 1, 2, 3, \dots, n - 2, n - 1\}$$

For representing our alphabet, we use a mod 26 system

$$Z_{26} = \{0, 1, 2, 3, \dots, 24, 25\}$$

And perform a one-to-one correspondence between the alphabet letters and the elements of this set.

Alphabet Assignment

$A \Leftrightarrow 0$	$K \Leftrightarrow 10$	$U \Leftrightarrow 20$
$B \Leftrightarrow 1$	$L \Leftrightarrow 11$	$V \Leftrightarrow 21$
$C \Leftrightarrow 2$	$M \Leftrightarrow 12$	$W \Leftrightarrow 22$
$D \Leftrightarrow 3$	$N \Leftrightarrow 13$	$X \Leftrightarrow 23$
$E \Leftrightarrow 4$	$O \Leftrightarrow 14$	$Y \Leftrightarrow 24$
$F \Leftrightarrow 5$	$P \Leftrightarrow 15$	$Z \Leftrightarrow 25$
$G \Leftrightarrow 6$	$Q \Leftrightarrow 16$	
$H \Leftrightarrow 7$	$R \Leftrightarrow 17$	
$I \Leftrightarrow 8$	$S \Leftrightarrow 18$	
$J \Leftrightarrow 9$	$T \Leftrightarrow 19$	

Shift Ciphers

If x is a numerical plaintext letter, we encipher x by computing the
$$y = (x + k) \bmod 26$$
, where k is in Z_{26}
Here y will be the numerical cipher plaintext letter.

***Note:** is called the key of the cipher and represents the shift amount.

EXAMPLE 16.10:

The Caesar cipher, developed by Julius Caesar is a shift cipher given by . Use the Caesar cipher to create a cipher alphabet. Then use it to encipher the message “RADFORD”. Note that the key

Solution:

To create the cipher alphabet, we substitute the mod 26 alphabet assignment numbers for each letter into the Caesar shift cipher formula and calculate the corresponding ciphertext letter numbers as follows:

$A \Rightarrow x = 0 \Rightarrow y = (0 + 3) \bmod 26 = 3 \Rightarrow D$
 $B \Rightarrow x = 1 \Rightarrow y = (1 + 3) \bmod 26 = 4 \Rightarrow E$
 $C \Rightarrow x = 2 \Rightarrow y = (2 + 3) \bmod 26 = 5 \Rightarrow F$
 $D \Rightarrow x = 3 \Rightarrow y = (3 + 3) \bmod 26 = 6 \Rightarrow G$
 $\vdots$
 $W \Rightarrow x = 22 \Rightarrow y = (22 + 3) \bmod 26 = 25 \Rightarrow Z$
 $X \Rightarrow x = 23 \Rightarrow y = (23 + 3) \bmod 26 = 0 \Rightarrow A$
 $Y \Rightarrow x = 24 \Rightarrow y = (24 + 3) \bmod 26 = 1 \Rightarrow B$
 $X \Rightarrow x = 23 \Rightarrow y = (23 + 3) \bmod 26 = 0 \Rightarrow C$

This gives the corresponding correspondence between the plain and ciphertext alphabets.

Plain	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Using the above table, we can encipher the message “RADFORD” as follows:

Plaintext	R	A	D	F	O	R	D
Ciphertext	U	D	G	I	R	U	G

Deciphering Shift Ciphers

Given that the key of the cipher k (the shift amount), a plaintext letter number x , and a ciphertext letter number y , we decipher as $x = (y - k) \bmod 26$

***Note:** In the deciphering shift cipher formula, $-k \bmod 26$ can be converted to its equivalent positive form by finding a positive remainder.

EXAMPLE 16.11: Supposed we received a ciphertext “YLUJLQLD” which was encrypted using a Caesar cipher (shift $3 = k$) . Decipher this message.

Solution:

Using the deciphering formula for shift cipher $x = (y - k) \bmod 26$, we have $x = (y - 3) \bmod 26$ (*). Note that since $-3 \bmod 26 = 23$ (this can be computed simply by taking $-3 + 26 = 23$), we can write the equation as $x = (y + 23) \bmod 26$ (**).

Either equations (*) or (**) can be used to decipher the message. We will use equation (**). Taking each letter of the cipher-text “YLUJLQLD” and using the mod 26 alphabet assignment, we obtain:

$$\begin{aligned}
 Y &\Rightarrow y = 24 \Rightarrow x = (24 + 23) \bmod 26 = 21 \Rightarrow V \\
 L &\Rightarrow y = 11 \Rightarrow x = (11 + 23) \bmod 26 = 8 \Rightarrow I \\
 U &\Rightarrow y = 20 \Rightarrow x = (20 + 23) \bmod 26 = 17 \Rightarrow R \\
 J &\Rightarrow y = 9 \Rightarrow x = (9 + 23) \bmod 26 = 6 \Rightarrow G \\
 L &\Rightarrow y = 11 \Rightarrow x = (11 + 23) \bmod 26 = 8 \Rightarrow I \\
 Q &\Rightarrow y = 16 \Rightarrow x = (16 + 23) \bmod 26 = 13 \Rightarrow N \\
 L &\Rightarrow y = 11 \Rightarrow x = (11 + 23) \bmod 26 = 8 \Rightarrow I \\
 D &\Rightarrow y = 3 \Rightarrow x = (3 + 23) \bmod 26 = 0 \Rightarrow A
 \end{aligned}$$

Hence, the plaintext is “VIRGINIA”.

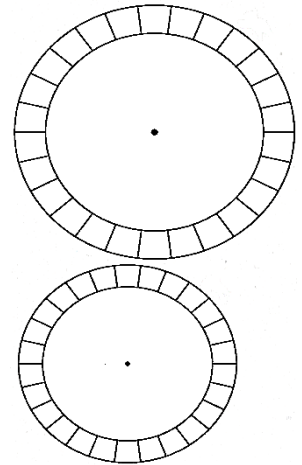
Learning Activity:

This activity will be done by 4 to 5 groups and do as follows:

- 1) Carefully cut around two circles.
- 2) Write the alphabet in BLUE around the small circles.
- 3) Write the alphabet in RED around the large circle.
- 4) Fix the small circle into the big circle using a paper fastener through the dot (mark with dot).
- 5) You are ready to use our Caesar Wheel. Remember that the plaintexts are written in BLUE while the ciphertext are in RED.

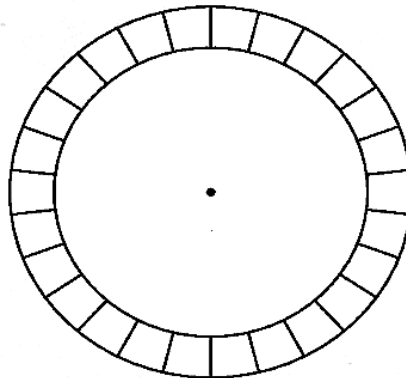
Enciphering = BLUE → RED

Deciphering = RED → BLUE



II. Using your Caesar Wheel solve the following problems:

- 1) Decipher the following messages:
 - a) **ltyufwnx** (Caesar cipher with shift +5)
 - b) **adiyevhznwjiy** (Caesar cipher with shift -5)
- 2) Encipher the following messages:
 - a) **hello tom** (Caesar cipher with shift +3)
 - b) **klondike nuggets** (Caesar cipher with shift +12)



Learning Evaluation:

- I. Determine whether the given number is a valid ISBN.
 1. 978 - 0 - 281 - 44268 - 5
 2. 978 - 1 - 55690 - 182 - 9
- II. Find the correct check digit of the given UPC.
 1. 0 - 79893 - 46500 - ? (Organics Honey)
 2. 8 - 88462 - 52148 - ? (Apple iPad Pro)

- III. Determine whether the given number is a valid credit card number.
1. 4416 - 5486 - 1785 - 6411
 2. 5591 - 4912 - 7644 - 1105
- VI. Encipher the following messages:
1. **three musketeers** (Ceasar cipher with shift +8).
 2. **Fly tonight** (Ceasar cipher with shift +6).
- V. Decipher the following messages:
1. **Yvibzm rdgg mjwdinji** (Ceasar cipher with shift +8).
 3. **Yvzam hk yebzam** (Ceasar cipher with shift +6).

References:

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